COMP 2711:
Discrete Mathematical Tools for Computer Science

What is discrete mathematics?

- Part of mathematics devoted to the study of discrete objects.
  - E.g., counting objects/events, analyzing processes (algorithms) with a finite set of steps.
- Topics in discrete mathematics
  - Logic
  - Counting (combinatorics)
  - Sets and their cardinality
  - (Discrete) probability theory
  - Number theory and cryptography
  - Induction and Recurrence
  - Algorithms and Asymptotic Notation
  - Graph theory

Why study discrete mathematics?

- It's a required course for CS students
  - Most important math for computer science
- Mathematical maturity
  - Modeling, reasoning, brain training
  - Basis for more advanced mathematics
- Useful in many other fields
  - Operations research (IELM)
  - Chemistry
  - Biology
  - Economics (game theory, decision theory, etc.)

Many interesting problems (Logic)



- Knights and Knaves
  - There are two kinds of inhabitants on an island, knights, who always tell the truth, and knaves, who always lie.
  - You encounter two people A and B.
  - A says: “B is a knight”
  - B says: “A is a knight”

  - Question: What you can say about A and B (knight/knave)?

Monty Hall Problem (Probability)

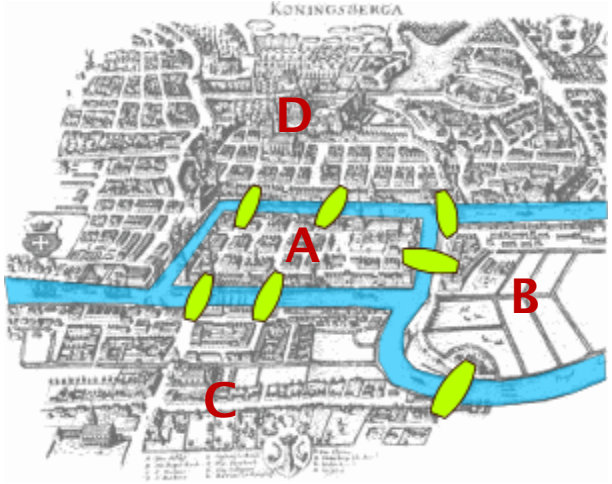
- Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a prize; behind the others, goats. You pick a door, say No. 1. The host, who knows what's behind the doors, opens another door, say No. 2, which has a goat. He then asks to you, “Do you want to switch to door No. 3?”

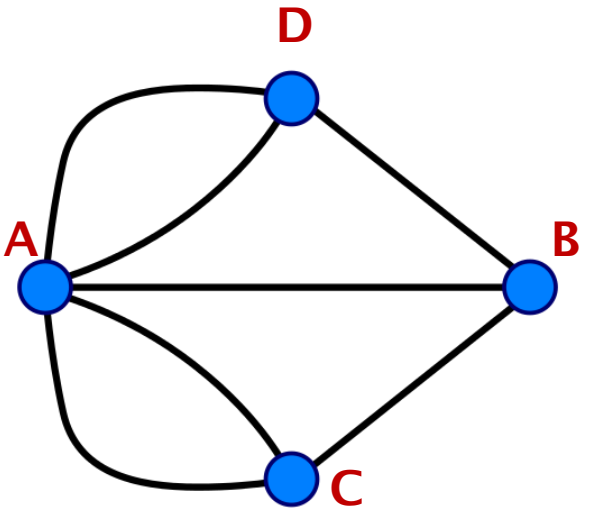
Answer 1: It’s 50-50, no need to switch.
Answer 2: Switch to door No 3.
87% respondents gave the wrong answer in a survey



Seven Bridges of Königsberg (Graphs)

- Q: Can you find a path to cross all seven bridges, each exactly once?





- Q: (Reformulated as a graph problem) Can you find a path in the graph that includes every edge exactly once?

How to Study Discrete Mathematics?

- Some students find it difficult
  - Many different topics
- Focus is on mathematical reasoning and problem solving
- Understanding more important than memorizing!
- Exercises!

Course Logistics

- We will try to keep the three sections synchronized with common assignments, tutorials and exams.
- No attendance requirements.
- Tutorials will start the second week of classes.
- Grading Scheme: Homework 20% (4 assignments), Mid-Term Exam 35% (30 Oct), Final Exam 45%.
- Textbook: Discrete Mathematics and its Applications. Kenneth H. Rosen, 7th or 8th Edition, McGraw-Hill.

L01: Propositional Logic

- Outline
  - Propositions
  - Compound Propositions
  - Propositional Equivalences

- Reading
  - Kenneth Rosen, section 1.1-1.3

Proposition

- **Definition**

  A **proposition** is a declarative statement (i.e., a sentence that declares a fact) that is either true or false, but not both.

- **Definition**

  The **truth value** of a proposition is true, denoted by T, if it is a true proposition, and false, denoted by F, if it is a false proposition.

- **Remark**

  Propositions are the basic building blocks of logic. The area of logic that deals with propositions is called **propositional logic.**

Examples

- Each of the following declarative statements is a proposition:
  - (a) Beijing is the capital of China
  - (b) COMP 2711 is an elective course for the COMP program
  - (c) $2 + 2 = 2^2$
  - (d) $1 + 1 = 3$

Propositions (a) and (c) are true but (b) and (d) are false.

Examples

- These statements are not propositions:

  (a) No parking

  (b) Who has an iPad?

  (c) $y = \log(x+1)$

  (d) $x^2 - 3x + 1 = 0$

  (c) and (d) can become propositions if x and y are assigned values.

- Logical paradoxes are not propositions

  (a) This sentence is false (Liar paradox)

  (b) All I know is that I know nothing (Socratic paradox)

Logical Operator and Truth Table

- **Propositional variable: We use lowercase letters p, q, r, … to represent propositions.**
- **Logical operators** or **logical connectives** can be used to turn existing propositions into new propositions.
- The definition of a logical operator can be given in the form of a **truth table** by enumerating all possible truth values of the proposition(s) involved.

Negation

- **Definition**
  Let p be a proposition. The **negation** of p, denoted by $\neg p$ or $\bar{p}$ and read as "NOT p", is the statement "it is not the case that p". The truth value of $\neg p$ is the opposite of the truth value of p.

p	$\neg p$
T	F
F	T

- Example:
  - p = "Tom's iPhone has at least 32GB of memory"
  - $\neg p$ = "It is not the case that Tom's iPhone has at least 32GB of memory"
  - $\neg p$ = "Tom's iPhone has less than 32GB of memory"

Conjunction

- **Definition**

  Let p and q be propositions. The **conjunction** of p and q, denoted by $p \wedge q$, is the proposition "p AND q".

  The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

- **Definition**

  The **disjunction** of p and q, denoted by $p \vee q$, is the proposition "p OR q".

  The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Exclusive Or

- **Definition**

  The **exclusive or** of p and q, denoted by $p \oplus q$, is the proposition that is true when exactly one of p and q is true and is false otherwise.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Exercise: Knight and Knave Puzzle 1

There are two kinds of inhabitants on an island, knights, who always tell the truth, and knaves, who always lie. You encounter two people A and B.

- A says: "B is a knight" (q)
- B says: "A is a knight" (p)

Question: What you can say about A and B (knight/knave)?

Lets define the propositions p :"A is a knight" and q:"B is a knight". Knights tell the truth, which means that their propositions are true. Knaves tell lies, which means that their propositions are false.

1] Assume that A is a knight (p is true). Then, according to his statement, q is true. Since B is a knight, his statement p is true, which is consistent with the assumption. Therefore, both A and B are knights ($p \wedge q$).

2] Assume that A is a knave (p is false). Then, according to his statement, q is false. Since B is a knave, his statement p is false, which is consistent with the assumption. Therefore, both A and B are knaves ($\neg p \wedge \neg q$).

Thus, I can conclude that A and B are of the same type (($p \wedge q$) $\vee$($\neg p \wedge \neg q$)).

Exercise: Knight and Knave Puzzle 2

- A says: "B is a knave" ($\neg q$)
- B says: "A is a knave" ($\neg p$)

Question: What you can say about A and B (knight/knave)?

p :"A is a knight", q:"B is a knight".

1] Assume that A is a knight (p is true). Then, q is false. Since B is a knave, his statement $\neg p$ is false, which is consistent with the assumption. Therefore, A is a knight and B is a knave ($p \wedge \neg q$).

2] Assume that A is a knave (p is false). Then q is true. Since B is a knight, his statement $\neg p$ is true, which is consistent with the assumption. Therefore, A is a knave and B is a knight ($\neg p \wedge q$).

Therefore, I can conclude that A and B are opposite types $(p \wedge \neg q) \vee (\neg p \wedge q) \equiv p \oplus q$.

Exercise: Knight and Knave Puzzle 3

- A says: "B is a knight" (q)
- B says: "The two of us are opposite types" ($p \oplus q$)

Question: What you can say about A and B (knight/knave)?

p :"A is a knight", q:"B is a knight".

1] Assume that A is a knight (p is true). Then q is true, and both A and B are knights. But this contradicts the statement of B that A and B are opposite types. Therefore, A cannot be a knight.

2] Assume that A is a knave (p is false). Then, q is false. Since B is a knave, his statement is false. Therefore, both A and B are knaves.

Conditional Statement

- **Definition**
  The **conditional statement** $p \rightarrow q$ is the proposition "if p, then q". Also called an **implication** (p implies q).
- p is called the **hypothesis** (or **premise**) and q is called the **conclusion** (or **consequence**).

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

- Note: There does not need to be any connection between the hypothesis and the conclusion.
  The true value of $p \rightarrow q$ depends only on the truth values of p and q.

Examples

- “If Beijing is the capital of China, then Beijing is a city of China”.
- True.
- “If Beijing is a city of China, then Beijing is the capital of China.”
- True.
- “If the moon is made of green cheese, then I have more money than Bill Gates. ”
- True.
- “If the moon is not made of green cheese, then I have more money than Bill Gates. ”
- False.

Necessary and sufficient conditions

- If $p \rightarrow q$ is true, then p is a **sufficient condition** of q, and q is a **necessary condition** of p
- Example:
  - "If I am elected, then taxes will be lowered."
  - "I am elected" is a sufficient condition for "lower taxes".
    - Other candidate may also lower taxes.
  - "Lower taxes" is a necessary condition for "I am elected"
    - If taxes are not lowered, then I am definitely not elected.

Other Ways to Express conditional statements in English

- Equivalent ways of expressing $p \rightarrow q$:
  - if p, then q
  - p implies q
  - p is a sufficient condition for q
  - p only if q (false q implies false p)
    - "The company can succeed only if it has sufficient funding."
  - q if p
  - q when p
  - q follows from p
  - q is a necessary condition for p

Converse, Inverse, Contrapositive

"If I am elected (p), then taxes will be lowered (q)."

- The **contrapositive** of $p \rightarrow q$ *is* $\neg q \rightarrow \neg p$.
  - if taxes are not lowered, then I am not elected
- The **converse** of $p \rightarrow q$ *is* $q \rightarrow p$.
  - if taxes are lowered, then I am elected
- The **inverse** of $p \rightarrow q$ *is* $\neg p \rightarrow \neg q$.
  - if I am not elected, then taxes will not be lowered
- $p \rightarrow q$ and its contrapositive are **equivalent**.

p	q	$p \rightarrow q$	$\neg p$	$\neg q$	$\neg q \rightarrow \neg p$
T	T	T	F	F	T
T	F	F	F	T	F
F	T	T	T	F	T
F	F	T	T	T	T

Biconditional Statement

- **Definition**

  The **biconditional statement** $p \leftrightarrow q$ is the proposition "p if and only if q".

  The statement is true when p and q have the same truth value and is false otherwise.

- Equivalent ways of expressing $p \leftrightarrow q$:
  - p if and only if q
  - p iff q
  - p is necessary and sufficient for q
  - if p then q, and conversely

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Ambiguity in natural language

- Sometimes, “if … then …” in natural language actually means “if and only if”.
- “If you finish your meal, then you can have dessert”
- What if you don’t finish your meal?
  - In natural language: then you can’t have dessert.
  - In logic: unspecified
- In this course, we should avoid such ambiguities.

Exercise: Differences between human and propositional logic

A real experiment was performed many years ago to study the relationship between human and propositional logic.

Human subjects were given the rule: "If the envelope is sealed, it must contain a stamp on the other side", and were asked to turn the minimum number of envelopes in order to verify the rule.

An alternative description: you work in the post-office and you want to make sure that **all sealed envelopes have stamps**, by turning the minimum number of envelopes. There are 4 cases:

1: You are shown a sealed envelope – do you need to turn it to check whether it contains a stamp?

Yes. I turn it over to make sure that it contains a stamp.

In terms of propositional logic you can think of

a] p as the proposition "the envelope is sealed",

b] q as the proposition "the envelope contains a stamp".

c] The rule is $p \rightarrow q$

Since p is true, you must check that q is also true. If $\neg q$, $p \rightarrow q$ is F.

Example of differences between human and propositional logic (cont)

2: You are shown an open envelope – do you need to turn it to check whether it contains a stamp?

No. The rule does not apply to open envelopes. They may contain a stamp or not. Since I have $\neg p$, $p \rightarrow q$ is T, and I do not care about q.

3: You are shown an envelope with a stamp – do you need to turn it to check whether it is sealed?

No. I do not care whether the envelope is sealed or open. Since I have q, I do not care about whether it is the case of p or $\neg p$.

4: You are shown an envelope without a stamp – do you need to turn it to check whether it is sealed?

Yes. The envelope must be open, otherwise there is a rule violation. Since I have $\neg q$, it must be $\neg p$ (**contrapositive:** $\neg q \rightarrow \neg p$).

The majority of human subjects gave the wrong answer for case 4.

Precedence of Logical Operators

- Multiple logical operators can be used to construct **compound propositions**.
- Parentheses can be used for clarity

  $$p \vee \neg q \rightarrow p \wedge q,$$

  may be written more clearly as

  $$(p \vee \neg q) \rightarrow (p \wedge q).$$

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

- Remark
  - Boolean operations in programming languages (e.g., ||, ^, !, == in C) correspond to logical operators
  - "**if** … **then** …" statements in programming languages do not correspond to $\rightarrow$

Tautology and Contradiction

- A **tautology** is a compound proposition that is always true (e.g. $p \vee \neg p$).
- A **contradiction** is a compound proposition that is always false (e.g. $p \wedge \neg p$).
- A **contingency** is a compound proposition that is neither a tautology nor a contradiction.

Tautology and Contradiction Examples

- $p \vee \neg p$ is a tautology
- $p \wedge \neg p$ is a contradiction
- $p \rightarrow \neg p$ is a contingency

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$	$p \rightarrow \neg p$
T	F	T	F	F
F	T	T	F	T

Logical Equivalence

- Compound propositions that always have the same truth values are logically equivalent.
- The compound propositions p and q are called **logically equivalent** if $p \leftrightarrow q$ is a tautology.
  - The notation is $p \equiv q$ (or $p \Leftrightarrow q$).
- Remark: The symbol $\equiv$ (and $\Leftrightarrow$) is not a logical operator and $p \equiv q$ is not a compound proposition, but rather is the statement that $p \leftrightarrow q$ is a tautology.

The use of ≡, ↔, =

- ≡ (⇔) denotes logical equivalence regardless of truth values of the propositional variables.
- ↔ is a binary logical operator.
- = is not used on propositions. It is only used between two mathematical objects.
- Examples:
  - $(p \rightarrow q) \leftrightarrow (\neg p \vee q) \equiv T$
  - $(p \leftrightarrow q) \equiv (p \rightarrow q \wedge q \rightarrow p)$
  - $(T \leftrightarrow F) \equiv F$
  - $(2 = 3) \wedge (5 \neq 4) \equiv F$
  - p: 2 + 2 = 4, p is true

Logical Equivalence: De Morgan's law

- Show that $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are equivalent

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$	$\neg(p \vee q) \leftrightarrow \neg p \wedge \neg q$
T	T	T	F	F	F	F	T
T	F	T	F	F	T	F	T
F	T	T	F	T	F	F	T
F	F	F	T	T	T	T	T

De Morgan's law examples

- $\neg(p \vee q) \equiv \neg p \wedge \neg q$
- Example: Express the negation of "Alice will send a secret message or Bob will send a secret message"
- p: Alice will send a secret message
- q: Bob will send a secret message
- By $\neg(p \vee q) \equiv \neg p \wedge \neg q$: "Alice will not send a secret message and Bob will not send a secret message."
- Another form of De Morgan law: $\neg(p \wedge q) \equiv \neg p \vee \neg q$
- Example: Express the negation of "today is Saturday and today is a holiday"
- p: today is Saturday
- q: today is a holiday
- By $\neg(p \wedge q) \equiv \neg p \vee \neg q$: "today is not Saturday or today is not a holiday"

Common Examples of Equivalences

1. Can we express $p \rightarrow q$ using negations and disjunctions?
   - Yes, $p \rightarrow q \equiv \neg p \vee q$ (Prove this using a truth table)
2. Can we express $p \oplus q$ using negations and disjunctions?
   - Yes, $p \oplus q \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$ (Prove this using a truth table)
3. Show that $\neg(p \rightarrow q) \equiv p \wedge \neg q$
   - $\neg(p \rightarrow q) \equiv \neg(\neg p \vee q)$ (by 1) $\equiv \neg(\neg p) \wedge (\neg q)$ (by De Morgan $\equiv p \wedge \neg q$

Exercise on Tautologies

- Is $p \vee (p \rightarrow q)$ a tautology?
- $p \vee (p \rightarrow q) \equiv p \vee (\neg p \vee q) \equiv (p \vee \neg p) \vee q \equiv \mathrm{T} \vee q \equiv \mathrm{T}$
- Tautology

- Is $p \vee (q \rightarrow p)$ a tautology?
- $p \vee (q \rightarrow p) \equiv p \vee (\neg q \vee p) \equiv (p \vee p) \vee \neg q \equiv p \vee \neg q$
- Not Tautology (when p is F and q is T, $p \vee (q \rightarrow p)$ is F).

Propositional Equivalences 1

- The following table summarizes the major propositional equivalences:

Equivalence	Name
$p \wedge \mathsf{T} \equiv p$ $p \vee \mathsf{F} \equiv p$	Identity laws
$p \vee \mathsf{T} \equiv \mathsf{T}$ $p \wedge \mathsf{F} \equiv \mathsf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws

Propositional Equivalences 2

Equivalence	Name
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathsf{T}$ $p \wedge \neg p \equiv \mathsf{F}$	Negation laws

Propositional Equivalences 3

Equivalence	Name
$p \to q \equiv \neg p \vee q$ $p \to q \equiv \neg q \to \neg p$ $(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$ $(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$ $(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$ $(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$	Involving conditional statements
$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$ $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$ $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$	Involving biconditional statements

Notes on Propositional Equivalences

- All laws (equivalences) van be proven using truth tables
- Sometimes you do not need the full truth table
  - Example of absorption law: $p \vee (p \wedge q) \equiv p$
  - When p is T(F), $p \vee (p \wedge q)$ is also T(F), irrespectively of the truth value of q
- Sometimes instead of truth tables, you can use other laws:
  - Example for implication law: $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$
  - $(p \rightarrow q) \wedge (p \rightarrow r) \equiv (\neg p \vee q) \wedge (\neg p \vee r) \equiv$
  - $\neg p \vee (q \wedge r)$ (distributive) $\equiv p \rightarrow (q \wedge r)$

Propositional Equivalence Example 1

Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences:

$\neg(p \vee (\neg p \wedge q))$ $\equiv \neg p \wedge (\neg(\neg p \wedge q))$ (De Morgan)

$\equiv \neg p \wedge (\neg(\neg p) \vee \neg q)$ (De Morgan)

$\equiv \neg p \wedge (p \vee \neg q)$ (Double Negation)

$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$ (Distributive)

$\equiv F \vee (\neg p \wedge \neg q)$ (Negation)

$\equiv \neg p \wedge \neg q$ (Identity)

Propositional Equivalence Example 2

Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology by developing a series of logical equivalences:

$(p \wedge q) \rightarrow (p \vee q) \equiv \neg (p \wedge q) \vee (p \vee q)$ (Equiv. def. of implication)

$\equiv (\neg p \vee \neg q) \vee (p \vee q)$ (DeMorgan)

$\equiv (\neg p \vee p) \vee (\neg q \vee q)$ (Associative)

$\equiv T \vee T \equiv T$

Note: You do not need to remember the names of the laws

Knights and Knaves Revisited

There are two kinds of inhabitants on an island, knights, who always tell the truth, and knaves, who always lie.

Let propositions *p be "A is knight" and q be "B is knight"*

You encounter two people A and B.

- A says: "We are both knaves": $\neg p \wedge \neg q$
- B says nothing

Question: What you can say about A and B (knight/knave)? Two cases:

1. If A's is a knight then both p, and $\neg p \wedge \neg q$ must be true. However, p and $\neg p \wedge \neg q$ cannot both be true at the same time (contradiction). Case 1 is not possible.

- If A's is a knave, then $\neg p \wedge \neg q$ is false, i.e.,$\neg$ ($\neg p \wedge \neg q$) is true. Using De Morgan's law,$\neg$ ($\neg p \wedge \neg q$) $\equiv p \vee q$. Since A is a knave, p is false. Therefore, q must be true. Conclusion: A is knave, B is knight.

Different Knights and Knaves Variant

You are on the island of knights and knaves and you want to find the way to the town. At a fork on the road, you have to decide whether to go left or right. There is a knight and a knave, but you do not know who is who. You chose one at random to ask for directions.

- What question should you ask, in order to find the correct way to the town.

Example: If you ask "Which is the direction to the town", you will get the correct direction from the knight and the wrong direction from the knave. You should ask a different question.

- Solution: "What the other person would answer if I asked which is the direction to the town"
- Explanation:

The knight would give the wrong direction, because this is what the knave would say.

The knave would also give the wrong direction, because he would reverse the knight's answer.

Thus, you should always go to the opposite direction of the answer.

Exercise: Logic Puzzle

- Determine the relative salaries of three coworkers using two facts.

1. If Fred is not the highest paid of the three, then Janice is.
2. If Janice is not the lowest paid, then Maggie is paid the most.

- Determine the relative salaries of Fred, Maggie, and Janice.

Notation: (F=1) denotes the proposition that Fred is the highest paid, (J=3) Janice is the lowest paid of the three etc.

1. I interpret the first fact as $\neg(F=1) \rightarrow (J=1) \equiv (F=1) \vee (J=1)$

   Either (F=1) or (J=1) is true (but not both since only one person can be the most highly paid)

2. I interpret the second fact as $\neg(J=3) \rightarrow (M=1) \equiv (J=3) \vee (M=1)$.

Assume that (J=1). Then the second fact cannot be true because $M \neq 1$ and $J \neq 3$.

Therefore, it must be that (F=1). For the second fact to be truth, it must hold that (J=3).

Thus, the (decreasing) order of salaries is Fred, Maggie and Janice.